

ActInf Textbook Group ~ Cohort 4 ~ Meeting 22 (Chapter 10 part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 22 (Chapter 10 part 2) | 56:45

All right. Welcome back.

Cohort 4.

This is our second discussion of Chapter 10.

Next week will be our final week overall.

So we'll talk about future textbook groups,

give a little time,

or people can fill out the forms now for feedback.

They can add feedback.

There's a lot of other pieces that we can talk about,

but this will be the last recorded conversation.

So nice times.

Let's go to 10.

And does anybody want to begin with any question

or we'll look at what's already been written?

So, you know, I have to ask questions.

Kind of general.

Since I'm new to this process,

one of the questions that I wish I had asked you sooner was,

and as we participate and contribute to,

and this process,

are there certain things that you would like to see?

That the authors would like to see?

So what I'm wondering about, like in terms of feedback that we did add through time is, you know, are they concerned about, you know, what audiences they're trying, attempting to reach?

Are they concerned about what principles, what concepts might go over people's heads and make them turn away completely collectively and friends prematurely?

So that's one question that's general.

The second question we have is, and I put, I added a row in the code.

I was thinking that if one of the objectives of this book is to have people who don't have the physics background walk away thinking, I would like to apply this to the work that I'm doing.

And of course, this is very selfish because that's basically describing itself.

So if, if we don't want to lose people, could there be a document somewhere?

And if it exists, let me know where it basically has a list of, okay, in order to understand,

to fully understand, and I'm just going to say this energy principle as an example, in order to fully understand the three energy principle, here is a pathway of knowledge that you need to complete.

And it goes in a sequence from, you know, concept A to concept N. And then you have the concept, equations, if, if the concept comes with equations.

And then just very succinctly described, you need to know this, and you need to know this, and then links to videos and resources online.

So that basically a person can follow a self-directed learning pathway.

And one of the things I've done over the past week is just, I decided I want to take a deep dive into the free energy principle.

Where did it come from?

Who came up with it?

How did they use it?

Do they use it in business, in engineering, et cetera?

And I have found this really helpful.

So that was my main topic for today.

Thanks.

Okay.

I'll give short responses to each and then anyone else can add.

So what, what, what, with the authors like Pard, Pazula, and Friston, I mean, they're happy that we're enlivening and, and working with the textbook.

We started doing it obviously like the first day that we could, and we've given them feedback on little typos and minor things, but also people have engaged with them on more substantial topics.

And they, as authors and as the publishing company, they're, they're at least open to our proposals.

There's no questions in the textbook.

And that one textbook isn't the be all end all.

So it's definitely, I think great to see how we can serve the authors and also the broader niche.

And in terms of like all backgrounds in the generalized learning curriculum, like this textbook group is low efforts.

All things considered, we don't have to prepare the materials.

And especially now having gone through it multiple times, we just show up and enjoy it.

But the broader project of making knowledge spaces so that people can have self-guided and have that be like across different languages and technical backgrounds.

I mean, that's the educational project that the Institute is working on.

So for that, we just need people who are seriously committed to actually engage in that.

So for that, at a slightly smaller scale, Oli, I believe in the resources page, this list of papers goes from, from less formal to more formal.

So, but this is an infinite regress.

You're going to find that you keep on pulling back to fundamental concepts, which may be good.

But yes, the textbook group is not the total educational paradigm.

It's just one offering.

It's just one offering.

And we eagerly look for people who want to stand up the textbook group and see what it can be.

All the low hanging fruit, like filling in the natural language descriptions of the equations and curating the discourse and adding more notes and all these things.

And then also the reach goals for the textbook group, like developing the software repositories that have the examples and all these other things.

And then even the broader educational goals that go outside of the textbook group.

Thank you.

Any, anyone else have a thought on that?

Yeah, Oli.

Yeah, so I believe being a relatively new discipline and even a new, I mean, research niche.

I don't think active inference has reached its full form yet in terms of being totally structured around a definite roadmap and so on.

So it's a bit like neuroscience at large.

So we know that unlike many other disciplines such as, I don't know, physics or mathematics, which have a relatively more structure to the knowledge base they have in such disciplines as neuroscience or an active inference or an active inference or many other cognitive science and psychological sciences as well.

We don't have this definite roadmap roadmaps yet.

So one way to approach active inference is to somehow determine the target topic we try to study using active inference framework.

And then try to find kind of rhizomatic or multi branch ways to approach or at least converge onto that topic.

So if we expect the learning path of active inference to be a totally linear and coherent pathway, getting us from point A to point B and beyond, I believe we might be disappointed at finding out that there isn't any such roadmap yet.

I mean, obviously in coming years, there might be more structure to the learning path of active inference.

But as it's typical for many other new disciplines and research areas, we're still exploring the forest and we don't have the cartography of the whole area yet at our disposal.

So I think that's a great point.

Thank you.

Yeah, that I may.

That makes sense.

And so I think that's a HIMO with a HIMO with a HIMO with a HIMO with a HIMO with a HIMO with a HIMO
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is in humans and the active inference framework is the first framework i have found that actually i can tell has the potential for allowing me to talk about things i couldn't even begin to talk about before but things i can actually give means to and um so in my case since i might ultimately have a very applied aspect to this which is i work on population health projects in education and hopefully soon in communities there's a certain urgency that i feel that i i can't go for example into an audience of public health scientists without being able to say something about why it's important to understand the freedom and that's what i'm yeah what i'm struggling with yeah we we're we're like making a thousand small tracers with people's questions here no questions no tracers a thousand questions a thousand tracers we can use curation and attention as well as language models to synthesize what we already have written into many of the bases now there's actually two components what you ask which is the necessary and sufficient information and the sequencing sequencing in linearization is a separate question than what are the fundamentals the fundamentals are what are covered in this textbook sanjeev's textbook the literature those are the topics that we return to again and again now they might be challenging which then begets the question of learner specific sequencing and context specific sequencing having the situational awareness to know what you need to go into or not when you're conveying active inference for a certain audience but yes we as part of a broader educational project outside of just the textbook group can use the ontology and other techniques to develop knowledge systems that that let us address these questions we're very very recently in the game and with very very few people months dedicated to the broader educational project but if it's something that people want to do there's always a niche for people who want to contribute to active inference education by doing something that's much like the textbook group like for cohort six or something or by diving back into the work that people have done over the years on the broader question like the fundamentals here that brock started um i guess a year ago or the other educational kind of first takes it's just a big journey but again with our knowledge engineering project and our literature compilations and all these other things it's the the challenge that we'll address yes i there are so many i'm finding all kinds of resources as you're talking thank you appreciate it any um thoughts on 10 or or any other questions

questions

well seen you asked about the computational complexity

as uh yes yes i did yeah yeah as far as i know the best ali please correct if we have a different information but i believe that the best information was the computational complexity right here

uh the computational complexity of uh model stream 6.1 branching time active inference paper here's 21 minutes in

and actually the computational complexity of of uh active inference is like it's almost what you'd expect is it just a hard problem or it's just uh

i'm i'm not able to compare it to np and sat and all of that but just at the within the um

computational resources space i mean it's okay it's the policy horizon here the time horizon depth times

um how many policies you have times how much time times how many hidden states there are okay just like how many things do you have to keep track of broadly and then branching time active inferences using heuristics to significantly reduce that complexity but they presented several complexity calculations in this paper and and with empirical run times yeah ali

i guess so in branching time active inference the attempt was to reduce the complexity class of the problems from uh exponential complexity to linear complexity uh so that was one way to do it but

uh a much more recent one actually it was from a few days ago uh is this recent paper by friston at

all namely supervised structure learning in which they have proposed an alternative uh way to uh reduce the

constrained mutual information of expected free energy so uh it's again another way to um reduce the

complexity uh class of the active inference or um uh probably any other uh inference based problems as well uh so yeah there are uh some attempts uh uh which um they explicitly address uh the problem of complexity of

um real real world active inference problems but uh one of the main criticisms of

uh uh or at least one of the main drawbacks of uh implementing uh active inference models in past years

was exactly uh based on uh the exponential complexity of uh of um higher level generative models and uh the ensuing

uh i mean uh intractability of the computation uh that resulted from that

yeah yes um so yeah so when i look at the uh generative model and the optimization problem it seems to me that it suffers the same uh curse of dimensionality uh as we uh had in dynamic programming when you so basically your time exponentially increase in the number of states uh the more number of states

whether it's heightened states or or explicit states and also the number of policies um that so if i

wonder what kind of a trick they have been able to use to to reduce the um computational complexity from exponential uh to um to kind of polynomial or linear you know linear is a form of polynomial right so

it's explanation is much much faster uh sorry it's much smaller than linear uh i mean near in terms of

the input you know so our input are um in in in in the um active inference model minimization problem

the the model misuse data uh which are the observations the uh dimensions of uh of states

uh the amount of policies uh the amount of policies yeah there's there's one issue i wonder if someone can help me so the the active inference defines the policy as as the um the action they take at each time

point affordance is the action you take at each time point policy is a sequence of actions for a given time horizon right right so it's a sequence of action uh at uh at each time point um so my understanding of policies that they are actually uh they they are not only a function of time not there are sequence of time but also there are sequence of states depends on what states you're in uh you take uh different actions actions not not just the time but also uh the hidden hidden states um so i wonder if they are the same um if what are the same you know so so uh action as a sequence of time versus action as a sequence of states states and time you know like ideally whether um you take the action at 2 p.m or 7 p.m or 11 p.m like the number of the time policy selection should be invariant unless the time is the determinant factor for which policy to take so the time number itself shouldn't matter and then as to whether a given observation or hidden state has a bearing on policy selection that's the question there's no reason to address that question in the abstract because obviously we can imagine situations that do and don't have that um and also while we're in the space of of computational accelerations like in the in the most recent message passing live stream now there's direct policy inference without expected free energy so i mean that's why we just hold on to our hats and stay in the game because it may not be required to iterate over all policies with an expected free energy compared the expected free energies you know do tree search on efe if you read the paper 56 and watch the discussions with magnus we heavily discuss this direct policy inference technique so sometimes the kind of all by all by all is like the pedagogical or just the first sketch but don't wait on the sideline because you're worried about a computational slowdown that may or may not happen either jump in and find out about that slowdown or jump in and then you'll find out that there's no slowdown or that by the time that you get to actually building that model in three months or six months that the computational environment has evolved but the general yeah

so i i'm actually i'm i'm uh supervising my student to build the um um you have the active influence model and when uh when i so so you have to you have to define uh the policies um

what it is you define you define the state you define the affordances and then the time horizon what is the furnace those are the actions that can be taken in a moment up down left right four affordances and then we're going to do you know two time steps and then there's 16 policies the policies are just the enumeration of all the possible sequences of affordances of that length so policy construction is trivial policy evaluation traditionally has been done with the efe but now there's an even more direct way to do it it doesn't mean that this is the best way to do it in every single person's situation for every single problem just that there is a technique now where you don't need to do expected free energy based comparison you can do generalized free energy based direct policy inference it just is that way

yeah so i'm repeating myself uh sorry just one last time hope somebody can help me so policies defined as function of time right so in the sense that you see it's a sequence of action as the policy defined in the textbook uh sequence of action and at time one what you do attempt to what you do left or right those actions but in general in my area in my uh dynamic programming area

the parties are defined as a function of state not not only time but also state uh in our example for example if whether it depends on what whether you're hungry or not hungry if one of the state of your senses uh data is that whether you're hungry or not then you know your policies it actually depends on whether you're going to go for food or or just just take a bite of some food depends on your state you're the hungry or not hungry state you know not necessarily it depends on time what yes in different areas of control terms like policy are used slightly differently but broadly we're talking about the same thing which is a policy is the sequence of actions and you can construct the model to be born upon by hunger or not so whether or not a given observation or hidden state has a bearing on a given policy selection that's up to the modeler that's not going to be addressed by active inference textbook group generalities but yeah if you make a model where hunger bears upon the choice to eat an apple then you have your answer if you do it where it doesn't you also have your answer okay but there isn't an in-principle answer to a situational design

so so butcher says that as um as a researcher i can define the policy in different way not not uh not just the sequence of action uh over time but a sequence of action over over states over um hidden six for example as part of constructing the entire generative model so a b c d e and so on you're right you decide not just the affordances and the time horizon which define the policy space but you also establish the sparsity and the connections of what else is included so if it's just a thermometer and then the thermometer is unrelated to um you know a slot machine player that's what you have but if the slot machine is conditioned on temperature that's what you have the slot machine is conditioned upon a taste sensor that's what you have so it's like a toolbox for us to take those steps and ask what if but we can't return to the framework and ask about whether certain factors influence each other or not again because we can just quickly think about examples where they would and wouldn't that's what chapter six is about is defining the whole generative model

not just we don't just hard code the policies

yeah thanks i i need to think a little more how to how to uh generalize the the policy you know to extend the model in chapter six uh to more general uh policy definition okay thank you anyway for uh can i think a little bit of a question um this um let's look at a few of the questions this is a great great starting one

section 10.4

there's a song lyric

and then this person has written

another verse from that song that's a little bit less

juvenile violence in theme

okay

anyone want to yeah yeah go ahead yeah i just wanted to i i need some clarification and i think it i don't know if it's relevant to uh launchers question um but

there are states hidden states that we attempt to infer from um our models

that's my understanding then there's perception

and then there's action and one of the things that um the themes in the book is that action and perception uh if i understood this correctly are seen as one and the same thing could you elaborate on that

a lot of great questions i you said that that through modeling like we we find or we discover hidden states a slightly more because we can't view them we can't see them right we have to infer them right we have a microcosm of the active agent which is that the active agent is only getting its sensorium the inbound and its its actions on the outbound and it is proposing latent unobserved hidden causes of sensory phenomena to bound its surprise

so um in in this kind when we write this kind of a model like figure 4.3 we put the hidden state down temperature in the room we put the observation down thermometer and then the agent does the inference on the hidden state so but but we constructed the hidden state space to be fixed the first or structure learning is like when the agent is going to be like rearranging combinations of hidden states

and and learning the structure of the world for itself um

action and perception are part of a unified

unity is plural at minimum two perception is not action simply now if you look across a blanket one agent's action becomes the other's perception so in that sense action and perception are just kind of like whose perspective are you taking from the blanket like you know screaming into the environment is the action from the yeller but then it's the perception from the environment so that's one sense but um action and perception are unified in active inference not by eradicating their differences but by understanding their complete articulation which is the two directions of traffic across the blanket so it's not that you know listening is the same as kicking your leg it's not that perception is action it's that they're part of a unified cognitive framework that has a unified imperative with free energy minimization unified handling of variables like this and so on

okay so having said that if you if having said that for one individual agent that when we move into really complex systems like the social insects you work on who

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For a complicated system like an ant's nest the system is the ant's nest can you tell me just plain english what's this is what are the actions examples of actions and examples of perceptions in that system

um sure well although the ant colony system is like the evolutionary replicator or or the kantian organism in the active inference model we made an agent-based model with an estimate and we used exactly the figure

4.3 type generative model right perception was pheromone and action was movements

so broadly the the selection of the perception action of the agent is similar to any other agent-based modeling technique if you modeled the ant colony as a single organism that's a totally cool and valid model there's always a plurality of models available so modeling a single colony as the agent maybe if you

wanted to model like a field site with just several thousand colonies and you didn't want to go into the nest mate level of detail or maybe the nest mate level is too coarse grains and maybe you need to do an organ specific model

these are maps not territories so it's whatever you need to do to take the next step on your path not about there only being like one ant's model

you need to do an ant's model right so then so then in in so as systems so systems to me are just i can't think of any system

that at least in the in with a phenomenon that i'm familiar with where there's anything an individual unit that's a very complex system so so it just seems to me that there are systems and nested systems within systems and interacting systems in the all kinds of systems interacting with each other etc so um so

i know there are different ways of thinking about complexity uh i think about it right now in a very qualitative way but it seems to me that

like for example in the ants that's nest you have all kinds of systems going you have the system of the you know maybe the uh the queen and everything that's associated with the reproduction then there are the systems of all the workers do you know keeping the group cleaning and so on and so forth so there are so many ways that you can take these concepts and and identify mark of mark of blankets that are relevant to different agents within systems and that can be then blown up to a whole system and in my understanding do this correctly yeah yeah i mean at the most zoomed out that's also true of linear modeling you could say well we could do a linear aggression of the queen's femur versus the number of seeds per minute per day squared you know there's no limit to the amount of linear aggressions um fmri analysis so there's just science and instrumentalism as a technique there's never any ending to how many models can be this is not active specific now you said there's nested interacting blankets or nested and interacting agents also very true we have enclosures of systems and laterally interacting collective behavior but those are expressions about the territory you're talking about how the colony really is you could choose to model it in different ways an active inference gives us a qualified way to stop that infinite regress by saying yeah there there are organs within the nest mate they do have a midgut and a brain and all this we're modeling this level and then that's the perfect stopping point for then a future study to pick up there and go one level deeper into the blanket

so it's not that one study must make all possible models and maps of all detail it's actually leading to a more principled way of collaborating across these different systems

that's the goal of the infra ants and and analogously for other systems i mean why do we have so many um square one models of ant foraging but there's never like sustained development of multiple labs with multiple species looking at different foraging environments because it's easy to make that custom model of that one ant in that one setting but then to generalize it to multiple ant species or multiple settings let alone tasks other than foraging it's very challenging so then the literature is basically littered with kind of toy models or intuition pumps but there's never a strong substrate to build a real

extensible model like a linux kernel for agent-based simulations

and and those have been tried but then those systems that try to generalize agent-based simulations themselves get mired down in technical details and and frictions of accessibility

but you've definitely identified the key

aspects of what we can explore

and that's why chapter six is provided chapter six doesn't say here's how nature is organized

chapter six says here's how you build that generative model and then how does chapter 10 end build the model and get surprised

it doesn't end with great now you've established how nature is

got it so so my next i go with my stupid question my next stupid question but i have to ask it

yeah because it seems like throughout the book there's like this this in some places it's very clear and

i've seen it in other contexts where it's stated clearly that what what the the goal is to to take

this understanding this deep structure based on the concepts and the architecture of the ecosystem of active influence whatever it looks like you know for whomever but there's kind of a consensus around

certain ideas that hold this conceptual framework together but the idea is that to be able to take

that and then go to into artificial intelligence and machine learning and say this stuff that we see in

living organisms this deep structure we can translate it into a completely artificial system and recreate

uh actions um by translating this deep structure into something that works in a machine um and then if

that's the case then i guess for me as a social scientist is the deep inference here that there is really no boundary between living and non-living systems from an active influence perspective

there's a lot to say on that um

one could debate whether the construction of a synthetic anything synthetic chemistry synthetic drawing a

synthetic active inference agent does constructing a synthetic anything legitimate any claim about the

territory just making an arts and crafts project doesn't change the nature of the trees and the sunset

so that's one point second point would be yeah instrumentally if not further we have a strong continuity

of diverse systems inert systems sleeping systems active systems and so on so from a instrumental

descriptive perspective we have an integrated modeling approach across different kinds of systems again

this doesn't mean life and non-life are the same exactly it means that there's a palette for people to say

this is what i mean by life and i made models that are on either side of the line and this one's in the gray zone

and someone can say well my line's over here well that's just the fact of the matter then that those people

have those different lines

yes okay thank you appreciate it yeah these are these are let's see if we can look at just just so that we can

okay i'll start at the bottom

okay yes thank you for this one address earlier yes join everyone is encouraged to join our broader

educational efforts and to enrich the textbook group where we actually will do these things

um which sections discuss is basing computation the most effective problem solving system for minimizing free energy not sure if in the book per se they might discuss bayesian optimality

um other people i know elsewhere take more like kind of first principles here's why things should be or

ought to be bayesian but suffice to say the framework doesn't rest upon bayesian computation being the most effective this is just a framework that we can use to describe any system now what's the most effective to actually do it that may or may not be doing bayesian statistics but the analytical formalism will never fail us as observers but that doesn't mean it's the best way to build it from the inside

it'd be helpful to provide definitions for thermodynamics free energy and of active variational free energy throughout the book free energy is used interchangeably with variational free energy

in this book they are probably only unless explicitly mentioned they're only talking about variational free energy they're never talking about gibbs unless they specifically called it out and i don't think they do they're only concerned with informational free energy not thermo

they're only concerned with the other people who are in the world but they're only concerned about the term of free energy

yeah yeah definitely there's always ways to connect it back and um let's go to the strongest alex keifer

psychophysical identity and free energy live stream 22. this is the strongest statement or one of the strongest statements made that that the thermodynamic free energy of a system directly encodes variational free energy so we're walking that pedagogical line you know they're different concepts you could link them how you want and then keifer just runs in and says no they're the same

or there are systems for which the thermodynamic free energy flows do entail variational free energy flows okay definition of brain

definitions for mind brain behavior yeah i mean there's there's the author's definition

and then there's our ontology and we meet every week to update the definitions and to translate them and to use them more coherently so people especially are interested in definitional questions let's do it um brains and centralized nervous systems

not exactly sure what you mean by the brain is a recent development

i mean

they're going back to the last common ancestor of vertebrates and invertebrates there was a centralized nervous system and it changed in structure and function even in the last tens of thousands of years

uh uh in that question it's kind of along the same lines of um the what's living and what isn't um

in this sense so i guess you know i would

you're right i mean it's it's it's about you as the observer and researcher decide where you're you know what what how you're going to define things what your affordances are what is the system that you're working with but i when i was first reading through the book i was just thinking are they

they this is only referring to systems where you have to have a nervous system and then what i was talking about

is that the the human brain in the in terms of the neurological variation that's out there in nature

it just has um you know um very different structure very complex structures

um anyway so it's along the lines so you know do you have to have this neurological kind of sensorium in order for to use active inference models and i think you guys are like a question good question

it absolutely does not require centralized nervous system or nervous cells it could be a pendulum
it could be in a nerve rock on the ground it doesn't need to be any kind of biological life form or any
kind of anatomy

thank you yeah that's really great but but but again the but most of the work has arisen from
carl friston and colleagues doing human neuroimaging again every single time in chapter five that's what
we emphasize chapter four generative model in essence chapter five they go to the case and the setting
where the most work empirically has been done which is human neuroscience but there's so many other
ways to go

thank you okay not is this a question or or what about this

i can't really see it's too small

opportunity to expand on niche construction i i agree with this

okay

yeah i mean the the um as everybody you know that sequencing question we talked about earlier
like the textbook is just one sequence and so some people are going to say connect it more to what has
empirically been done and i want to understand how brains work i want to have that first person driver's
seat perspective i want the stories to be about me picking up the water me getting out of the chair me
listening to music other people are going to be about bigger than humans or different than humans
shouldn't we use only the technical mathematical

aren't um appeals to personal experience possibly even misleading

everyone's right that's why we need a bigger knowledge system that can generate those sequenced paths
so if people want the human 101 no math in this language they can have that path and if somebody wants a
different language different technical level no human examples that is also going to be fully consistent
um and so is is um so chris field's uh course on information on physics as information processing
is that you know in this context um trying to connect dots here um is that um a way of
saying or his perspective is we don't have to make the works on complicated for at the quantum from
the quantum physics perspective we can actually think about active influence in that different way
is in my mind and trying to get at the deeper structure of information processing yeah this this textbook
was written probably largely between 2018 and 2022 maybe even 2021 because it was published in march
2022

and it was centered around fristen's 2019 particular physics um monograph developed a lot
the quantum free energy principle work with fields glaze brook levin marciano at all
only was in 2022 live stream 40. so this textbook does not include any quantum informatics chris fields and
colleagues with their backgrounds in the quantum information sciences made that connection between free
energy

energy principle and quantum so that connection is basically unaddressed in this textbook

and quantum so that connection is basically unaddressed in this textbook

uh only only only in the most minimal possible way hinting at the bayesian mechanics

so this textbook is um

um i mean

those who love the history and philosophy of science again let us reconvene to write these histories
but this book is in some ways like a coda on the 2010s of active inference because already in the 2020s
we're heading off in some new directions and so it's kind of like an inflection point
that consolidated a huge amount of the past work but now with message passing and julia on this side and
quantum fvp on that side and far more empirical applications happening all around
it's a different environment for doing um active inference learning and application and for writing a
textbook as we could ask sanjeev about so this textbook's not everything and again just from a chapter 10
perspective in only about 200 pages 200 very short pages
what they've done is incredible and that's why we can always return to it and and um
yeah even though it's only been 18 months since it's being released
but this is the
discussion we've had that does not conclude cohort four i hope that people return um next week
to to discuss more and um yeah update things develop things add projects add questions develop the
answers
will be used and will be used and will be used and will be used and will be used and will be used and
Appreciate it.
See you next time.
See you next time.
Thank you, Daniel.
Bye-bye.
Bye-bye.